

Slide 1: Title Slide

Good afternoon, everyone. I am Arnob Chakroborty, Student ID: M.Sc 260227. Today, I am going to present my seminar project titled **“Malware Detection Using Artificial Intelligence”**. This presentation focuses on how AI techniques are transforming malware detection in modern cyber security.

Slide 2: Abstract

Malware has become one of the most serious cyber security threats, causing financial losses and data breaches worldwide. Traditional methods like signature-based detection are no longer effective against advanced threats such as polymorphic and zero-day malware. Artificial Intelligence, especially Machine Learning and Deep Learning, provides a smarter solution by learning patterns and detecting unknown threats. This work surveys different AI techniques, feature engineering methods, datasets, evaluation metrics, and real-world applications. It also highlights challenges like adversarial attacks and discusses future directions such as Explainable AI and Federated Learning.

Slide 3: List of Contents

This presentation is organized into several sections: Introduction, Background, Types of Malware, Literature Review, Machine Learning and Deep Learning techniques, Hybrid Framework, System Architecture, Results, Case Study, Challenges, and Conclusion. I will explain these sections one by one.

Slide 4: Introduction

Malware is a rapidly evolving threat affecting individuals, organizations, and governments. It includes viruses, ransom ware, spyware, and Trojans, all designed to damage systems or steal data. Traditional detection methods rely on known signatures, which makes them ineffective against new or unknown attacks. Artificial Intelligence improves detection by analyzing behavior and identifying unusual patterns in real time. This makes AI a powerful and adaptive tool in cyber security.

Slide 5: Background

There are three main traditional detection approaches:

- **Signature-based detection:** Matches files with known malware signatures.
- **Heuristic methods:** Use rules to identify suspicious behavior.
- **Behavior-based systems:** Monitor system activity in real time to detect anomalies.

Among these, behavior-based systems are more effective against unknown threats.

Slide 6: Types of Malware

There are several types of malware:

- **Virus:** Attaches to files and spreads when executed
- **Trojan Horse:** Disguises as legitimate software
- **Worm:** Spreads automatically across networks
- **Ransom ware:** Encrypts data and demands payment
- **Spyware:** Steals sensitive information
- **Adware:** Displays unwanted ads
- **Rootkits:** Provide hidden unauthorized access

Each type has a different attack mechanism and impact.

Slide 7: Literature Review

Several researchers have contributed to this field:

- Kolter and Maloof used machine learning for malware detection
- Saxe and Berlin applied deep neural networks
- Raff proposed using raw executable files in deep learning

Other systems like DREBIN and DL4MD improved Android malware detection.

Overall, research shows:

- Machine Learning improves detection
 - Deep Learning performs even better
 - Hybrid approaches combining both are the most effective
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Slide 8: Machine Learning Techniques

Machine Learning techniques are divided into:

- **Supervised Learning:** Uses labeled data for training
- **Unsupervised Learning:** Finds patterns without labels

Common algorithms include Decision Trees, SVM, and Random Forest. These methods depend heavily on feature extraction. This table shows the key differences among the machine learning techniques.

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Slide 9: Deep Learning Techniques

Deep Learning automatically extracts features and handles large datasets. Popular models include:

- Artificial Neural Networks, Long-Short Term Memory
- Convolutional Neural Networks
- Recurrent Neural Networks, Generative Adversarial Networks.

These models generally outperform traditional machine learning in accuracy.

Slide 10: AI-based Hybrid Framework

A hybrid framework combines both Machine Learning and Deep Learning. It uses both **static analysis** (code inspection) and **dynamic analysis** (runtime behavior). This approach improves accuracy and reduces false positives. For this framework, the models that are used are ANN, CNN, SVM, and Random Forest.

Slide 11: System Architecture

Here, the datasets are EMBER, virus-share, CICIDS and Microsoft Malware Dataset. And this is the workflow of the methodology. The system consists of several modules:

1. **Data Collection:** Gathers files, network traffic, and logs
2. **Preprocessing:** Cleans and normalizes data
3. **Feature Extraction:** Extracts API calls, opcodes, and metadata
4. **Model Training:** Trains ML and DL models with AI
5. **Detection Engine:** Classifies files as benign or malicious
6. **Response Module:** Takes actions like blocking or alerting

This pipeline ensures efficient malware detection.

Slide 12: Results and Discussions

The results show that AI-based models outperform traditional methods.

Key observations:

- Deep Learning achieves higher accuracy
- Neural networks improve detection performance
- Hybrid models reduce false positives

Performance is evaluated using metrics like Accuracy, Precision, Recall, F1-score, and ROC-AUC. Here, this graph shows the comparison of all the models. Here, we can see that Hybrid model outperforms all the models.

Slide 13: Case Study

A case study on ransom ware detection shows:

- Data was collected and features were extracted
- A supervised learning model was trained

- The system successfully detected ransom ware
- Data encryption was prevented

This demonstrates the real-world effectiveness of AI-based detection.

Slide 14: Limitations & Challenges

Despite its advantages, AI-based malware detection faces challenges:

- Data imbalance
- High computational cost
- Limited datasets
- Generalization issues

These challenges must be addressed for practical implementation.

Slide 15: Conclusion and Future Work

In conclusion, AI significantly improves malware detection by identifying patterns and detecting unknown threats. However, issues like false positives and high computational cost remain. Future research focuses on:

- **Explainable AI (XAI):** Making models more transparent
- **Federated Learning:** Improving privacy
- **Edge AI:** Enabling real-time detection on devices

These technologies will shape the future of cyber security.

Slide 16: Bibliography

This work is based on several research papers and studies from journals and conferences related to AI and cyber security. These are the papers that I have used for my research.

Slide 17: Thank You

Thank you for your kind attention, everyone.
